

Francis Alvarez

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Research Manager | Data Science | Program Evaluation

Data and research analytics leader with 6+ years of experience applying statistical modeling, forecasting, machine learning, and program evaluation techniques to statewide transportation programs serving tens of thousands of participants and informing multimillion-dollar policy decisions. Experienced leading cross-functional teams and translating complex analyses into executive-level recommendations that influence policy and program design.

Education

San Diego State University, Master of Science in Applied Mathematics

University of California, San Diego, Bachelor of Science in Applied Mathematics

Experience

Managing Research Analyst, Center for Sustainable Energy *Mar. 2025 – Present*

- Direct program evaluation activities for California's Driving Clean Assistance Program, overseeing survey research, forecasting, dashboard development, and stakeholder reporting to support statewide electric vehicle adoption initiatives.
- Led analysis of electric pickup truck adoption for California's Clean Vehicle Rebate Project (CVRP), producing public-facing reports and executive presentations that informed program strategy and stakeholder decision-making
- Managed and mentored a team of three junior analysts, overseeing analytical quality, project execution, and professional development.
- Produced a logistic regression analysis to describe which sociodemographic features best predict reliance on statewide incentives to purchase electric vehicles.

Senior Research Analyst, Center for Sustainable Energy *Feb. 2023 – Mar. 2025*

- Co-authored a [whitepaper](#) explaining the differences in EV consumer behavior by residence type, which was presented at the [Electric Vehicle Symposium 36](#) and the Mobilize Summit 2023.
- Served as the research lead for statewide programs, including Louisiana's NEVI [Program](#), Power New Mexico's Transportation Electrification Program, and Washington State's EV Charging [Program](#). Led client meetings, coordinated team efforts, and ensured projects were delivered on time and within budget.
- Developed forecasting models of California's electric pickup truck market to estimate program participation and quantify the multimillion-dollar financial impact of proposed incentive structures, informing statewide program planning and policy decisions.

Research Analyst, Center for Sustainable Energy *July 2020 – Feb. 2023*

- Developed and [presented](#) multi-scenario financial forecasts to estimate the impact of various design changes to the CVRP incentive program.
- Co-authored a white paper analyzing trends between the CVRP program and statewide new vehicle registration data to inform policy and program design.
- Developed and maintained a public-facing dashboards tracking statewide EV rebate participation and survey outcomes, providing stakeholders with program performance metrics.

Data Analyst, Wondersign *Apr. 2019 – July 2020*

- Developed a data pipeline in Python to standardize data feeds from multiple furniture brands into a single unified format.
- Developed machine learning classification models that improved product labeling accuracy across tens of thousands of furniture catalog records.

- Designed a Google Data Studio dashboard to track data quality trends across partner furniture brands to enforce data consistency standards from clients.

Skills

Programming & Analytics: Python, SQL, R, MATLAB

Statistical & Research Methods: Statistical Modeling, Forecasting, Logistic Regression, Survey Analysis, Program Evaluation, Machine Learning, Segmentation Analysis

Data Visualization & Reporting: Tableau, Excel, Dashboard Development, Data Visualization, Data Storytelling

Leadership & Domain Expertise: Project Management, Stakeholder Management, Cross-Functional Leadership, Technical Writing, Public Speaking, Consumer Research, Market Research, Policy Analysis

Tools: Git, Jira, Teamwork

Languages

English (native), Spanish (proficient)